

FINAL SMART CONTRACT SECURITY & TESTNET VALIDATION REPORT

VoltPay VLT - BNB Smart Chain

FINAL TESTNET VERDICT: PASS

Technical evidence record covering source integrity, automated analysis, local and fork testing, live BSC Testnet transactions, anti-whale behavior, fee processing, ownership finalization and source verification.

Report date	4 August 2026
Report ID	VLT-SEC-TV-2026-08-04
Scope	Frozen VoltPayV2 FINAL source and completed BSC Testnet validation
Network	BNB Smart Chain Testnet - chainId 97

This document is a technical validation record. It is not a guarantee of future security, investment performance, legal compliance, or an independent third-party manual audit certificate.

1. Executive Summary

The frozen VoltPayV2 FINAL source completed the planned validation program without any observed Critical, High, or Medium severity finding in the recorded SolidityScan result. Local Hardhat tests, a BSC mainnet-fork test, live testnet trading, fee processing, anti-whale enforcement, irreversible configuration actions, post-renounce treasury processing, and source verification were all exercised.

Test result: 93/93 local Hardhat tests passed plus 1/1 BSC mainnet-fork test passed, for a recorded total of 94/94 PASS.

Live testnet result: BUY fee 2% PASS, SELL fee 4% PASS, processFees PASS, treasury payment PASS, removeLimits PASS, anti-whale reverts before limit removal PASS, large transaction after removal PASS, ownership renounce PASS, and post-renounce treasury fee processing PASS.

Verification result: the primary testnet deployment was verified by Sourcify and displayed by BscScan Testnet as Source Code Verified - Exact Match with Solidity 0.8.35, optimizer 200 runs, and Paris EVM settings.

Severity	Count	Interpretation
Critical	0	No findings
High	0	No findings
Medium	0	No findings
Low	4	Missing Events - 4 instances
Informational	18	Non-exploitable observations / code-quality signals
Gas	85	Gas optimization observations

Important interpretation of '107': SolidityScan's 107 total detector instances are the arithmetic sum of 4 Low + 18 Informational + 85 Gas observations. The number must not be interpreted as 107 exploitable security vulnerabilities.

2. Frozen Source Identity and Build Configuration

The source used for the completed validation cycle was treated as frozen. No Solidity modification was introduced during the recorded testnet validation sequence.

Final source file	VoltPayV2_FINAL_0_8_35.sol
SHA-256	9d708e7e874bdf72533d1f59d976049dfeeff9f3555aeccfd4c2bfdc94cbcdade
SPDX	MIT
Solidity compiler	0.8.35 / 0.8.35+commit.47b9dedd
Optimizer	Enabled - 200 runs
EVM version	Paris
Hardhat	3.12.0
Node.js	24.18.1
OpenZeppelin package	5.6.1
Token supply	200,000,000 VLT - 18 decimals

Integrity control: the same SHA-256 value was rechecked across the final master source and the working testnet copy before deployment/verification activities.

BscScan's source view showed the SPDX line `// SPDX-License-Identifier: MIT`. A separate top-card license metadata field appeared as '-NA-' in the explorer UI; this was treated as explorer metadata/display behavior because the verified source itself contains the MIT identifier.

3. Validation Methodology and Automated Analysis

The validation program combined multiple complementary layers rather than relying on a single scanner score.

Layer	Tool / Environment	Recorded result
Local unit/integration tests	Hardhat	93/93 PASS
Fork test	BSC mainnet fork / real PancakeSwap V2 behavior	1/1 PASS
Symbolic/security analysis	Mythril 0.24.8	No issues reported
Static analysis	Slither	Findings reviewed; no code change required for the frozen FINAL
Automated contract scan	SolidityScan QuickScan	93.18/100 - GREAT
Live-chain validation	BSC Testnet	Two deployments used for primary and dedicated anti-whale proof
Source verification	Sourcify + BscScan Testnet	Verified / Exact Match

SolidityScan detail: Critical 0, High 0, Medium 0, Low 4 (one category: Missing Events, 4 instances), Informational 18, Gas 85. The free QuickScan did not expose the exact four source locations behind the Low category without an upgraded report.

Scope limitation: automated tools can produce false positives, false negatives, style observations, and gas suggestions. Their output was therefore interpreted together with executable tests and live-chain behavior.

4. Executable Test Coverage

The 94/94 recorded test result covered the following functional and security-relevant areas:

1. Deployment metadata, total supply, ownership, treasury and router configuration.
2. Pair creation and initial trading-disabled state.
3. Initial fee, transaction-limit, wallet-limit and swap settings.
4. Trading enablement and owner authorization controls.
5. Buy and sell fee logic.
6. Anti-whale maxTx and maxWallet behavior.
7. Swap threshold / maximum swap amount logic and liquidity/treasury split.
8. Configuration finalization and irreversible lock behaviors.
9. Limit removal behavior.
10. AMM-pair recognition, including the Stage 13 unregistered AMM-pair bypass case fixed in RC6.
11. Fee processing, treasury accounting and post-renounce operational path.
12. Real PancakeSwap V2 interaction under a BSC mainnet fork.

Stage 13 hardening: the previously identified unregistered AMM-pair bypass was corrected in RC6 and then covered by tests for buy fee, sell fee, maxTx buy and maxTx sell behavior.

5. Primary BSC Testnet Deployment

Token	0x33F1b91EEDB8058C105d8E625c61bc2d36C7C9C0
Pair	0x3C116B8b57Ac96D9E94f0e8c718DFeDbD2371c24
Deployer	0x3d1d7fc44539AA407f83CA11A2f779f2bbC33a96
Treasury	0x58E841fD39a6247AE43919Bd4E1d4CfaAe704D6C
Buyer	0xBf871EC0e725Bf6E843f04825eD5898A133597Fe
Deploy transaction	0xec4d7a662e309de4998c30820ffda023619bf737f754a5e01ce858d6dccad515
Deploy block	123111873
Deploy gas	5,210,084
Recorded deploy cost	0.0005210084 tBNB

Post-deployment checks confirmed the expected owner, router, treasury, pair bytecode, 200,000,000 VLT total supply, full initial supply at the deployer, and trading disabled.

Initial live parameters: buyFee 200 bps, sellFee 400 bps, maxTx 30,000 VLT, maxWallet 30,000 VLT, swapThreshold 12,000 VLT, maxSwapAmount 30,000 VLT, pendingTreasuryBNB 0, configFinalized=false, feesLocked=false, limitsRemoved=false, swapSettingsLocked=false.

6. Live Liquidity, Trading and Fee Validation

Action	Transaction	Observed result
Approve 3M VLT for router	0xe897514feea9c9376e226b6e42cf23fddb9c11a3866f00cb01dd93df158e2923	PASS
Add LP: 3,000,000 VLT + 0.05 tBNB	0x3ce008ba6d96d0caf1182262b8b36b9c7dbda493ccf541d8a1ac2a76062fde7c	PASS
Enable trading	0x7ba88c5a4ee3bb5925c591ee7ce4e74390e0033100835feb0c4548b613ad0758	PASS
BUY 0.0001 tBNB	0xff9e24810c7d0eb387e082ba4b17867611acd3caed57a420a9c754d848106586	Gross 5,976.071760765512037972 VLT; fee 119.521435215310240759 VLT = exact 2%
SELL 3,000 VLT	0x984ea1148d270de90e1dcef725a2a69186d3f8134ee1a5e1f58edde9d88c3d6a	Pair received 2,880 VLT; fee 120 VLT = exact 4%
processFees	0x3cc27229c445f718fb04dee362f390d393dc8d7e419da35e4a7b24f5efc63842	Treasury pending BNB created; DEAD LP increased
retryTreasuryPayment	0x99eb6a0ddd1ad57e9753b7e459d0d5c08632e821af7469a5f8ddb969deb82fe1	Pending treasury BNB -> 0; exact payment received

Fee processing evidence: before the first processFees call the contract held 239.521435215310240759 VLT. The call generated 0.000001996030332382 tBNB pending for treasury and increased DEAD LP by 7,723,036,215,323,836 units. retryTreasuryPayment then reduced pending to zero and treasury received the exact pending amount.

7. Irreversible Configuration and Post-Renounce Operation

Operation	Transaction	Result
finalizeConfig	0x8c7ddb43e8ee0489807291ef3577a0a61bc3067f46a170e7e0b742afca7f73d8	PASS
lockFees	0xca2494208e7ac3b7e86a759591c815d08df563387302a5d8e162f692b6f171d1	PASS
lockSwapSettings	0x607b5d0c3b4532a949ec9be620e13100041a9dc6ef4f2d6bf06df0d0a3b0cde1	PASS
removeLimits	0x1ed52e77e2851ccadd3dd431cb37c0b5b8866721d3b64565e449580acd093ef5	PASS
Large BUY after removeLimits	0xb43220389f8726c5ff396bf8cf3dd5885d93b4f114e9c0cdc8d74be03e002cba	Gross 58,594.885480299244393037 VLT > old maxTx; final buyer wallet > old maxWallet; exact 2% fee
renounceOwnership	0x1f44c8ca240c8c134fcacb5cdb113641582aee8969352a5cfa9a598811556c82	Owner -> zero address; former owner unable to transfer ownership
processFees after renounce	0xa87bc77f49049da24d7f9e540b635daca813a44dc8c0320d78fbc05beace4de9	PASS; treasury pending created; DEAD LP increased
Treasury payment after renounce	0xdac5f44e00e837fe19292762b4a1d538b2e5559c00cb572e85a36d0c9810ddb7	PASS; pending -> 0; contract BNB -> 0

Operational conclusion: the dedicated treasury remained able to execute the intended fee-processing/payment path after ownership was renounced. Owner-only recovery functions are, by design, unavailable after renounce.

8. Dedicated Live Anti-Whale Proof

Dedicated token	0x462cDE6bf0afe975Fc634F0782Bf28b74cB15C30
Dedicated pair	0x13F722c2Ff8CdEF8F994681671Fd1d8ac0fa47e3
Deploy TX	0x1d36a6b0ac5185a57cf9f91a88992848865da24cc65bfd9f58bb49a596751168
LP TX	0xb1a16f9bcf75d254a250a0714c49ddd1929d275146486a6751451a13bb383ecb
Enable trading TX	0x4e062cf3de1aed09e0abde43cae09ed6fdf883991f0238b0f204f132aae25fd2

Scenario	Transaction	Live result
BUY gross > 30,000 VLT before removeLimits	0xfc63aed29dee4f7f24378a9026e4d2137967a5d84ca629027730cc4b8ad29926	REVERT - PASS; buyer balance remained 0
First BUY below maxTx	0xb9cac3aebca52347306f100e5d59076c19aff2ec43639c16f33394eb061a6739	Success; buyer 17,499.930416665009920595 VLT
Second BUY: each tx < maxTx but projected wallet > 30,000	0x5f4439f7b92131697dc4f4a700837317b1f306e83933359c8f54c3291c514f43	REVERT - PASS; balance unchanged
SELL 31,000 VLT before removeLimits	0x1b7918922f9585ba55f408e1de99ec40ae0297c6a8972e87178807b11f09cc07	REVERT - PASS; balance and allowance unchanged
removeLimits	0x29d0736e666e5eb3da9ae118c6da992d7888be0279f8c5dce399d304fb4ced20	On-chain state confirmed limitsRemoved=true
Same 31,000 VLT SELL after removeLimits	0xdb7e860d41930fa43e061af32ebbc551b762fbc8845ca07da076a66eead39241	PASS; pair received 29,760 VLT; contract fee 1,240 VLT = exact 4%

Conclusion: live anti-whale limits blocked oversized buy, wallet-cap overflow, and oversized sell while limits were active; after removeLimits, the same oversized sell succeeded and the 4% sell fee remained correct.

9. Source Verification and External Scanner Record

Service	Status	Evidence / interpretation
Sourcify	PASS	Contract verified successfully for chainId 97. Minimal compiler input first failed; Hardhat automatically retried full solc input and verification succeeded.
BscScan Testnet	PASS - Exact Match	Source Code Verified; contract VoltPayV2; compiler v0.8.35+commit.47b9dedd; optimizer 200; EVM Paris.
SolidityScan QuickScan	93.18/100 - GREAT	Critical 0 / High 0 / Medium 0 / Low 4 / Informational 18 / Gas 85.
GoPlus token_security	Deferred	Supported-chains response included BSC mainnet 56 but not BSC Testnet 97.
Quick Intel	Deferred	UI network selector exposed BSC mainnet but no BSC Testnet entry.

Public verification references:

Sourcify: <https://sourcify.dev/server/repo-ui/97/0x33F1b91EEDB8058C105d8E625c61bc2d36C7C9C0>

BscScan Testnet source: <https://testnet.bscscan.com/address/0x33F1b91EEDB8058C105d8E625c61bc2d36C7C9C0#code>

GoPlus and Quick Intel deferral is a network-support limitation for this testnet phase and is not a contract finding. Both belong in the post-mainnet-deployment external-check checklist.

10. RPC Consistency Note and Final On-Chain Snapshot

Two testnet scripts initially reported false negatives after successful transactions because a public load-balanced RPC returned stale 'latest' state immediately after the receipt. Later receipt/state checks confirmed the transactions and final state were correct.

Mainnet hardening recommendation: after a transaction receipt, wait for confirmations and verify critical state either at the receipt block or against a reliable RPC endpoint instead of trusting an immediate unpinned 'latest' read.

owner	0x00
tradingEnabled	true
buyFee	200 bps (2%)
sellFee	400 bps (4%)
configFinalized	true
feesLocked	true
swapSettingsLocked	true
limitsRemoved	true
Contract VLT	0.498535867058123637
pendingTreasuryBNB	0 tBNB
Contract BNB	0 tBNB
Treasury tBNB	0.032151829047249203
Pair reserve0	2939719963367889480573973
Pair reserve1	51039797464274316
DEAD LP	46264413402505495

11. Mainnet Readiness - Controlled Next Phase

The completed testnet program supports proceeding to a controlled mainnet preparation phase. It does not mean mainnet deployment has already occurred.

01. Re-verify FINAL SHA-256 immediately before mainnet compile/deploy.
02. Freshly verify official PancakeSwap V2 mainnet router, factory and WBNB addresses against official sources and on-chain code.
03. Use separate dedicated mainnet deployer and treasury wallets; never expose private keys or seed phrases in scripts, chat, reports or shell history.
04. Confirm BSC mainnet chainId 56, RPC endpoint, deployer address and BNB balance before any write.
05. Compile with Solidity 0.8.35, optimizer 200, EVM Paris and compare artifact/source identity.
06. Run preflight/gas simulation before sending the deployment transaction.
07. Deploy FINAL, then immediately verify owner/router/treasury/pair/supply/trading=false.
08. Verify source on BscScan and Sourcify before opening trading.
09. Configure all intended settings before finalizeConfig or any other irreversible lock.
10. Add real liquidity and verify pair reserves / LP token ownership.
11. Enable trading only after configuration review; execute one small real BUY and one small real SELL and verify 2% / 4% fees.
12. Remove limits only at the chosen launch moment after observation; verify post-removal behavior.
13. Operate processFees with quote/slippage safeguards and post-receipt state confirmation.
14. Perform GoPlus, Quick Intel and other mainnet external checks after the real contract address exists.
15. Treat renounceOwnership as a separate manual decision after stable operation - never bundle it into the launch script.
16. Lock/prove LP according to the launch policy and publish verifiable evidence.

12. Final Testnet Validation Verdict

Control	Result
Source integrity	PASS - frozen FINAL identified by SHA-256
Local executable tests	PASS - 93/93
BSC mainnet-fork test	PASS - 1/1
Combined executable tests	PASS - 94/94
Mythril	PASS - no issues reported
SolidityScan severity gate	PASS - 0 Critical / 0 High / 0 Medium
Primary BSC Testnet deployment	PASS
Live BUY / SELL fee validation	PASS - exact 2% / 4%
Fee processing / treasury	PASS
Anti-whale live proof	PASS
Irreversible configuration actions	PASS
Post-renounce treasury path	PASS
Sourcify verification	PASS
BscScan source verification	PASS - Exact Match
Final on-chain snapshot	PASS

FINAL TESTNET VERDICT: PASS. Based on the evidence recorded in this report, the planned testnet validation program for the frozen VoltPayV2 FINAL source is complete. The next phase is controlled mainnet preparation and deployment, with fresh infrastructure verification and separate operational approvals for irreversible actions.

Report boundary: this document records the completed testnet validation state as of 4 August 2026. Any future source-code change creates a new artifact and requires re-validation.